Blockchain based voting application

ABSTRACT

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The objective of this engineering thesis was to develop a web application for conducting online elections leveraging blockchain technology. The project utilizes a hybrid architecture that integrates a traditional client-server web application with the decentralized Ethereum network via smart contracts.

From the user's perspective, the system provides a streamlined voting process that eliminates the need for account registration or specialized knowledge of cryptocurrencies and blockchain mechanics. To ensure accessibility, the voting procedure is free of charge for the voters, achieved through the implementation of a transaction intermediary. A key feature of the application is a system that generates a unique private key locally within the user's web browser. This key is never transmitted to external servers, ensuring that only the voter has access to it for signing ballots, which are subsequently recorded on the blockchain.

The application was designed with a rigorous focus on election integrity and personal data protection. To confirm voter identity, a multi-stage verification procedure was implemented, consisting of preliminary authentication using a PESEL number and a system-generated code, followed by a challenge-response protocol. Upon successful verification, a time-limited session is created, during which the user can cast their vote.

Practical tests were conducted to confirm connectivity and functionality on the Sepolia testnet, where the election results were recorded. Furthermore, automation tools were developed for efficient user data management and email communication. To ensure transparency, the system includes dedicated commands for extracting blockchain data, enabling a fundamental audit of each election. Future development of the application will focus on transitioning to a public mainnet, alongside continuous improvements in system security and reliability.